



Introduction to IP & Addressing



Internet Protocol

The IP in TCP/IP

- IP is the network layer protocol
 - packet delivery service (host-to-host).
 - translation between different data-link protocols.

IP Datagrams

- IP provides connectionless, unreliable delivery of *IP datagrams*.
 - Connectionless: each datagram is independent of all others.
 - Unreliable: there is no guarantee that datagrams are delivered correctly or even delivered at all.

- Version
- IP header length (HLEN)
- Type-of-service
- Total length
- Identification
- Flags
- Fragment offset
- Time-to-live
- Protocol
- Header checksum
- Source address
- Destination address
- Options
- Padding
- Data

0	4	8	16	19	24	31
VERS		HELEN	Service Type	Total Lenth		
Identification			Flags	Fragment Offset		
Time to Live		Protocol	Header Checksum			
Source IP Address						
Destination IP Address						
IP Options (if any)					Padding	
Data						
...						



IP Addresses

- IP addresses are not the same as the underlying data-link (MAC) addresses.

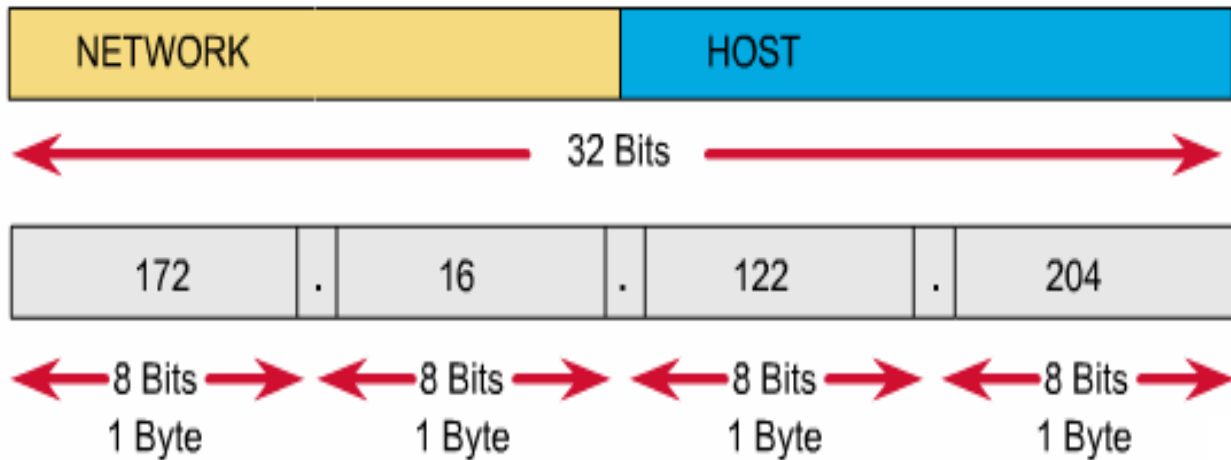
IP Addresses

- IP is a network layer - it must be capable of providing communication between hosts on different kinds of networks (different data-link implementations).
- The address must include information about what *network* the receiving host is on. This is what makes routing feasible.

IP Addresses

- IP addresses are *logical* addresses (not physical)
- 32 bits.  IPv4 (*version 4*)
- Includes a network ID and a host ID.
- Every host must have a unique IP address.
- IP addresses are assigned by ICANN
(*Internet Corporation for Assigned Names and Numbers*).

IP Address as a 32-Bit Binary Number



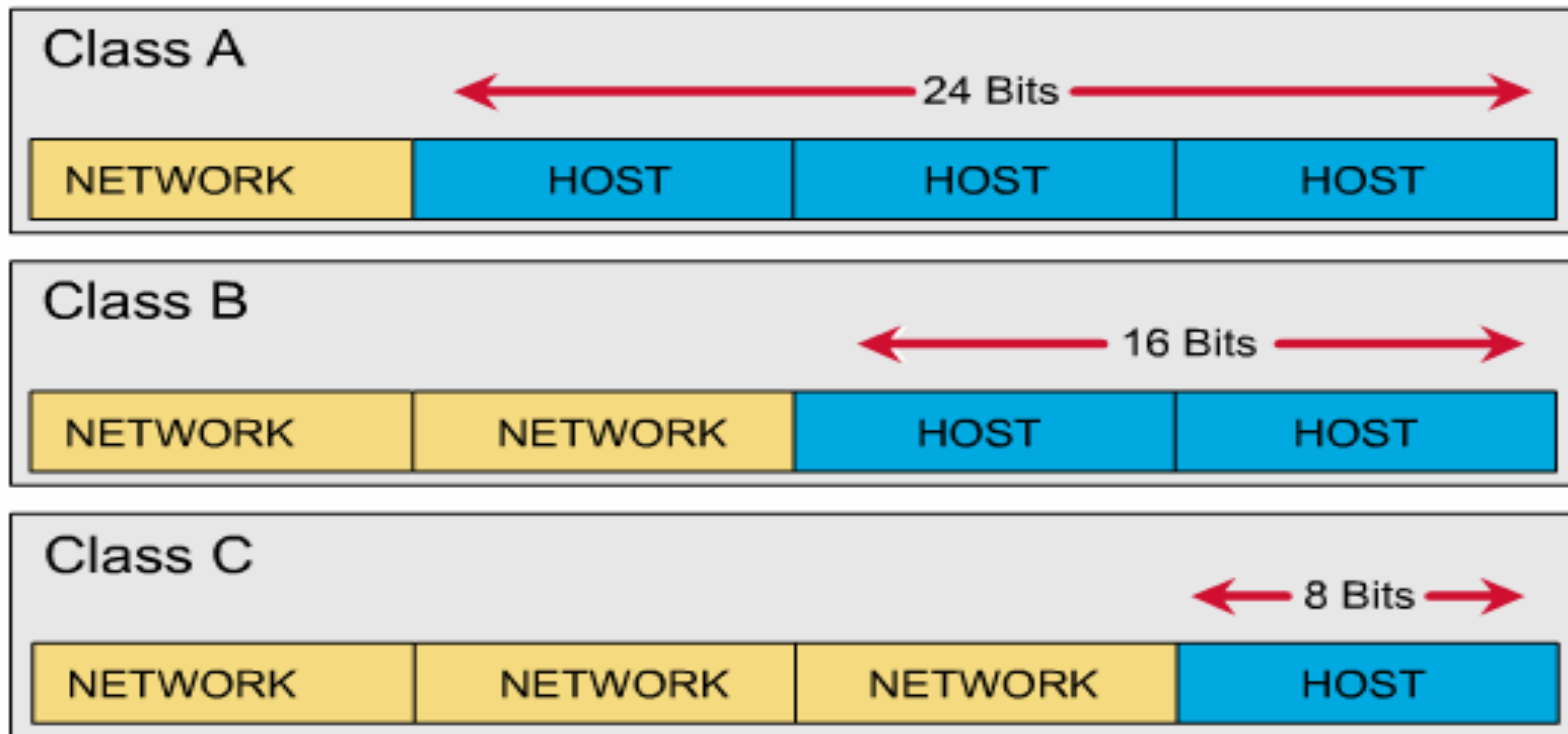
Binary and Decimal Conversion

$2^{(7)}$	$2^{(6)}$	$2^{(5)}$	$2^{(4)}$	$2^{(3)}$	$2^{(2)}$	$2^{(1)}$	$2^{(0)}$
128	64	32	16	8	4	2	1

192.57.30.224

11000000.00111001.00011110.11100000

Classes of Network IP Addresses



IP Addresses as Decimal Numbers

# Bits	1	7	24
--------	---	---	----

Class A:

0	NETWORK#	HOST#
---	----------	-------

# Bits	1	1	14	16
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Class B:

1	0	NETWORK#	HOST#
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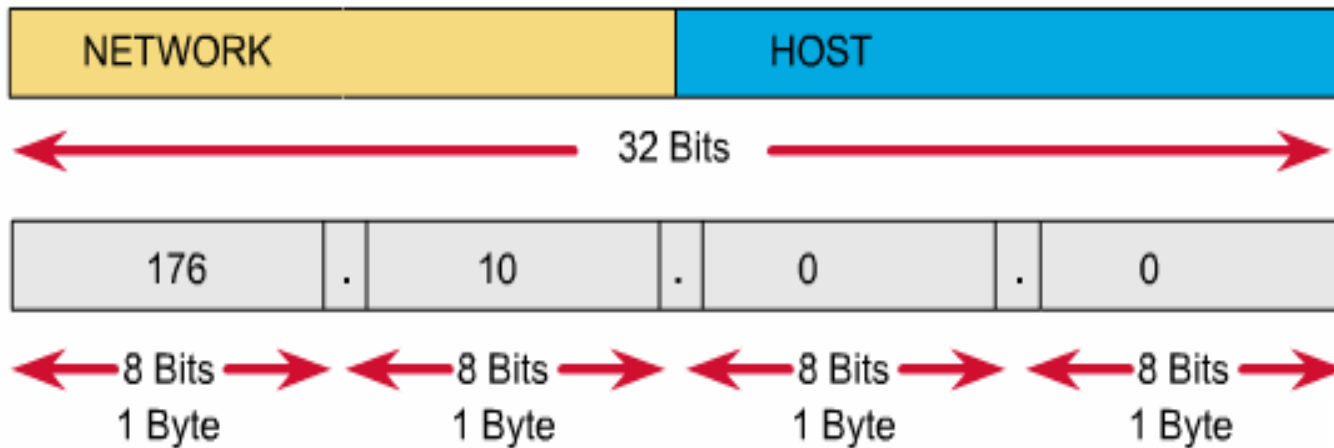
# Bits	1	1	1	21	8
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Class C:

1	1	0	NETWORK#	HOST#
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Network IDs and Broadcast Addresses

An IP address such as 176.10.0.0 that has all binary 0s in the host bit positions is reserved for the network address.



An IP address such as 176.10.255.255 that has all binary 1s in the host bit positions is reserved for the broadcast address.

Private Addresses

The following ranges are available for private addressing

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

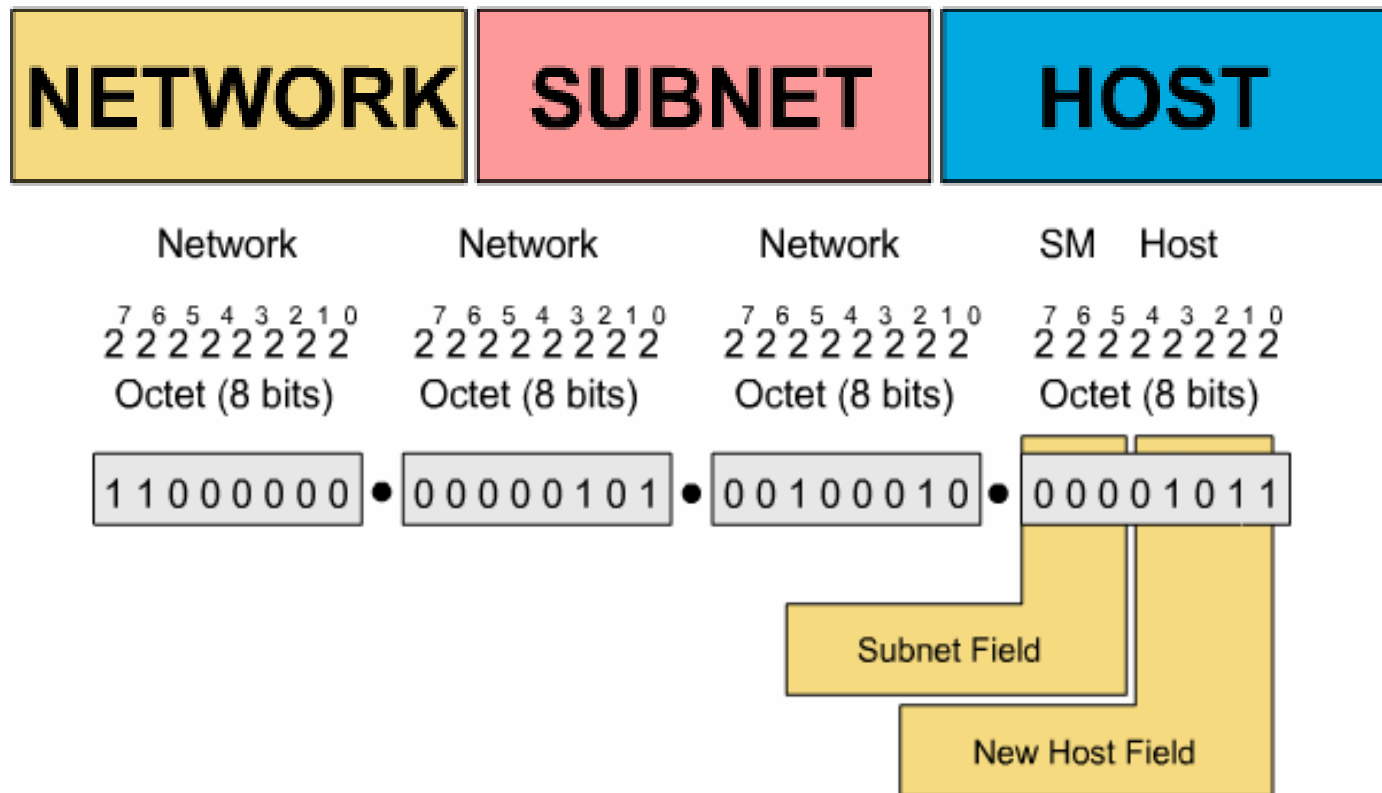


Subnetworks

To create a subnet address, a network administrator borrows bits from the original host portion and designates them as the subnet field.

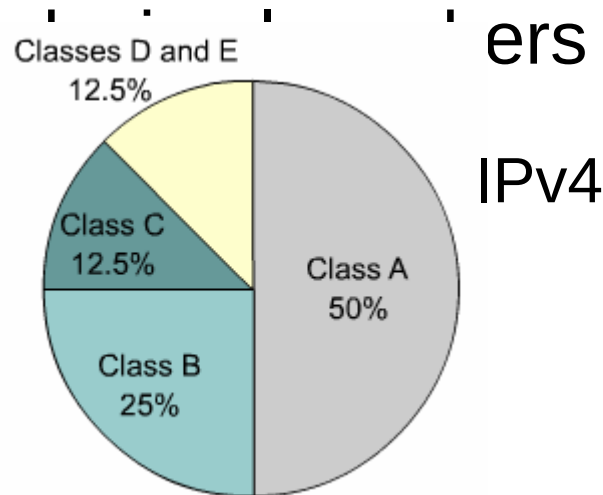
Subnetworks

SOLUTION: Create another section in the IP address called the subnet.



IPv4 versus IPv6

- IP version 6 (IPv6) has been defined and developed.
- IPv6 uses 128 bits rather than the 32 bits currently used in IPv4.
- IPv6 uses hexadecimal characters to represent the 128 bits.





Obtaining an IP Address

■ Static addressing

- ☐ Each individual device must be configured with an IP address.

■ Dynamic addressing

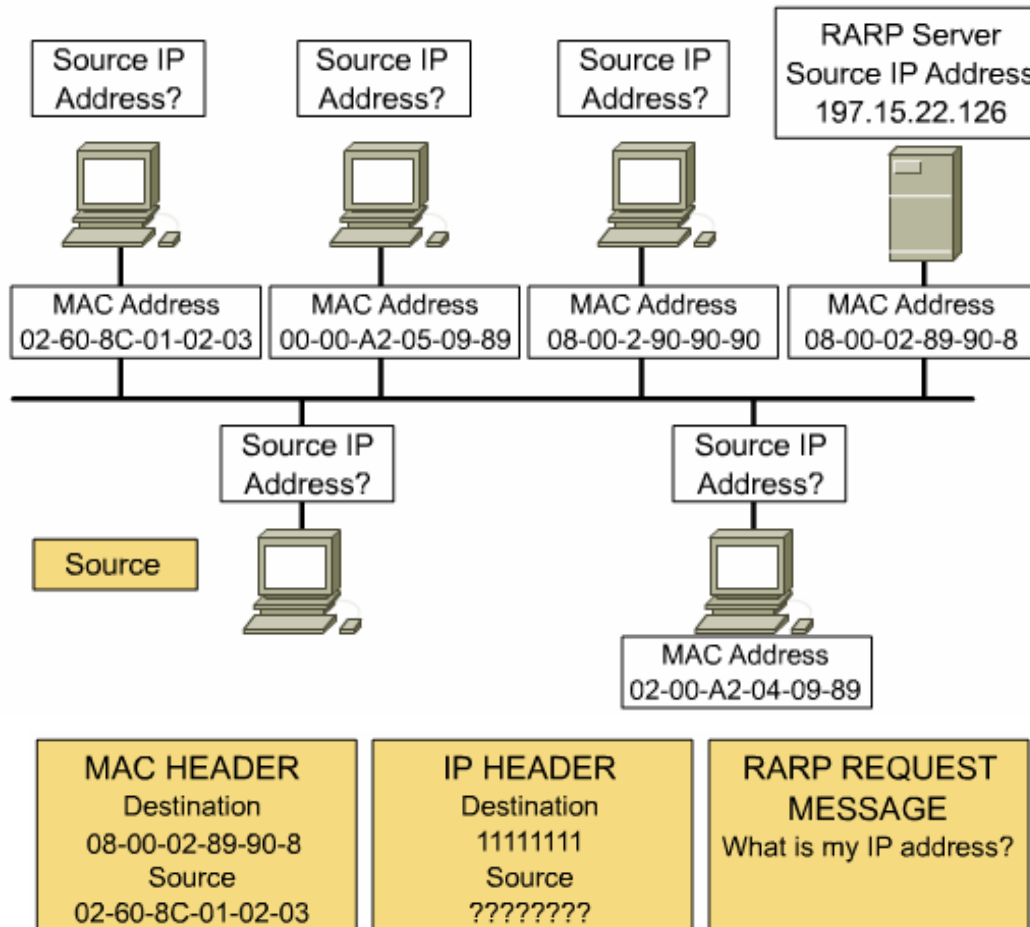
- ☐ Reverse Address Resolution Protocol (RARP)
- ☐ Bootstrap Protocol (BOOTP)
- ☐ Dynamic Host Configuration Protocol (DHCP)
- ☐ DHCP initialization sequence
- ☐ Function of the Address Resolution Protocol
- ☐ ARP operation within a subnet



Static Assignment of IP Addresses

- Each individual device must be configured with an IP address.

Reverse Address Resolution Protocol (RARP)



The source initiates a RARP request, which helps it detect its own IP address.



BOOTP IP

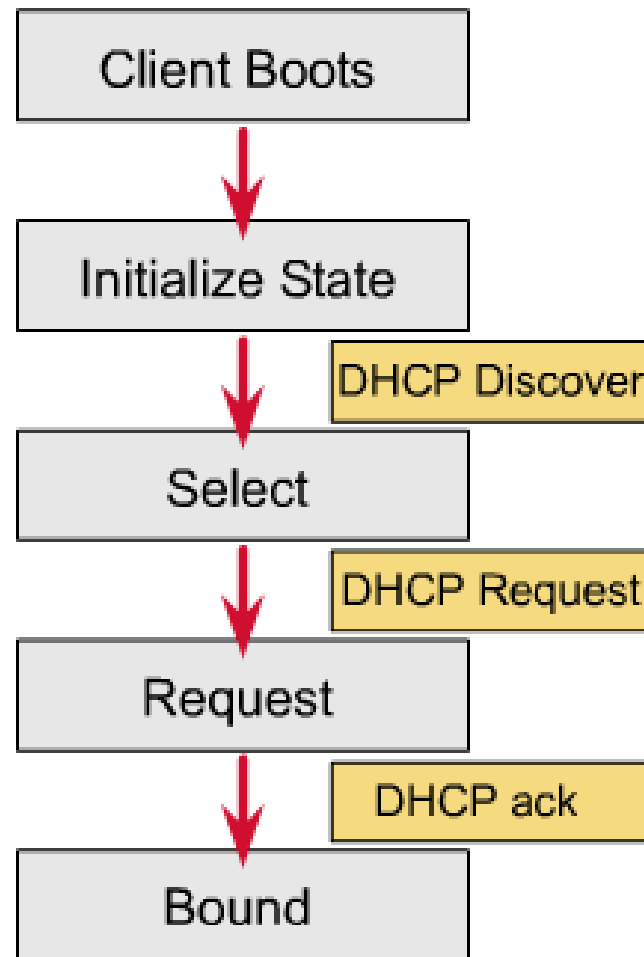
- The Bootstrap Protocol (BOOTP) operates in a client/server environment and only requires a single packet exchange to obtain IP information.
- BOOTP packets can include the IP address, as well as the address of a router, the address of a server, and vendor-specific information.



Dynamic Host Configuration Protocol

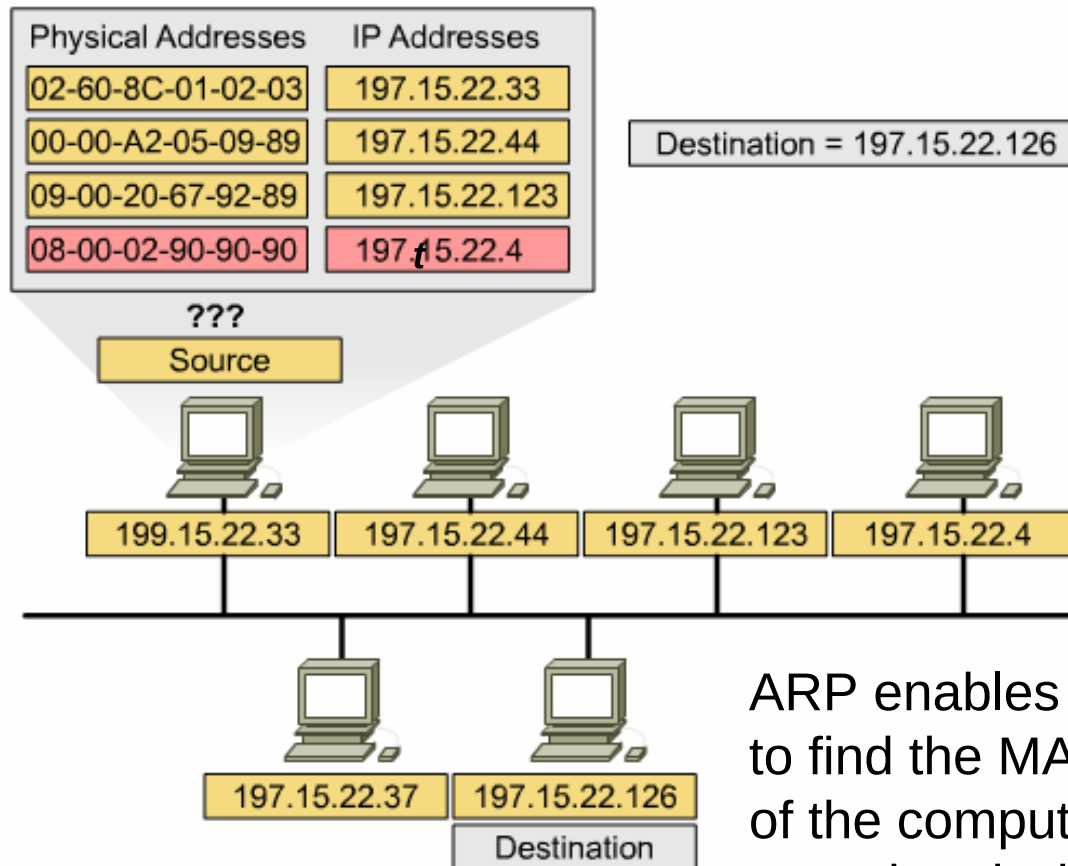
- Allows a host to obtain an IP address using a defined range of IP addresses on a DHCP server.
- As hosts come online, contact the DHCP server, and request an address.

DHCP Initialization Sequence



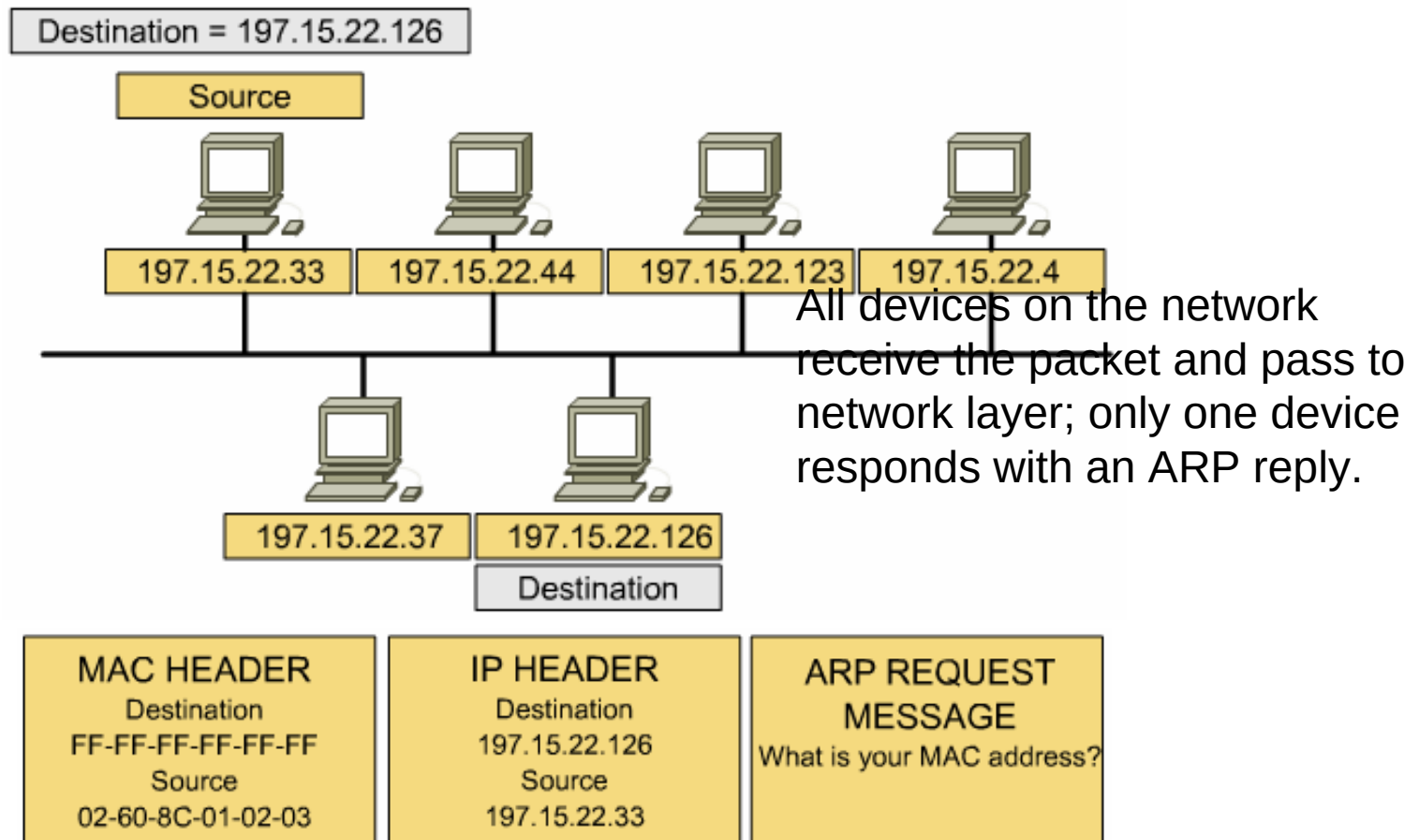
Client collects
DHCP offer
responses from
the server.

ARP

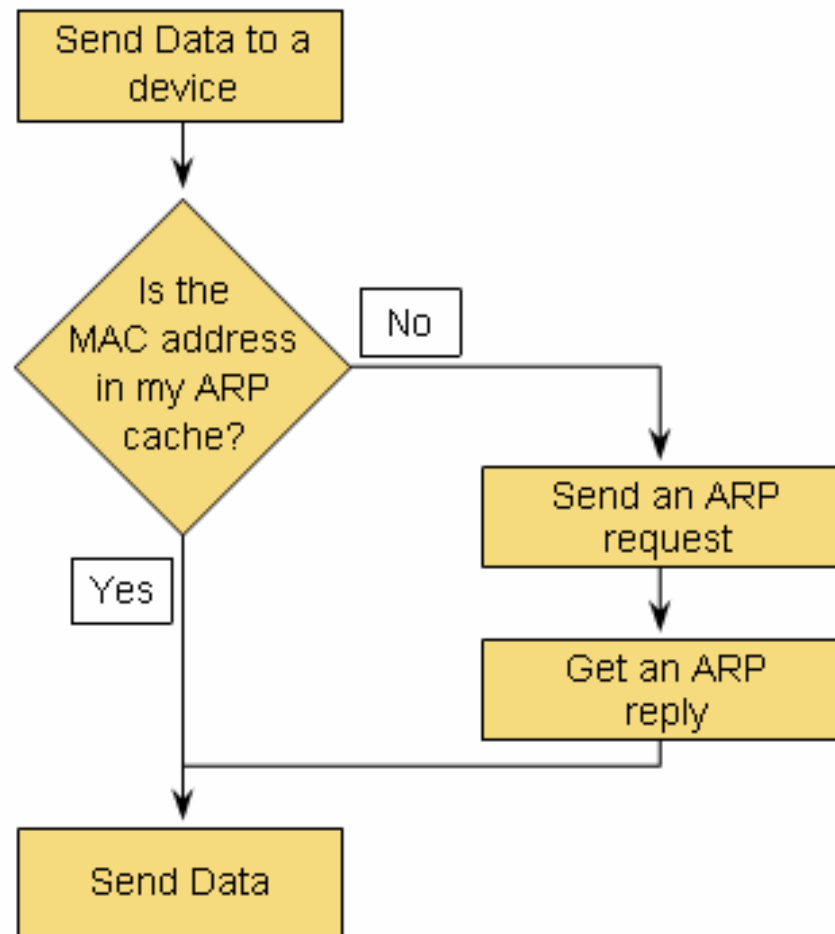


ARP enables a computer to find the MAC address of the computer that is associated with an IP address.

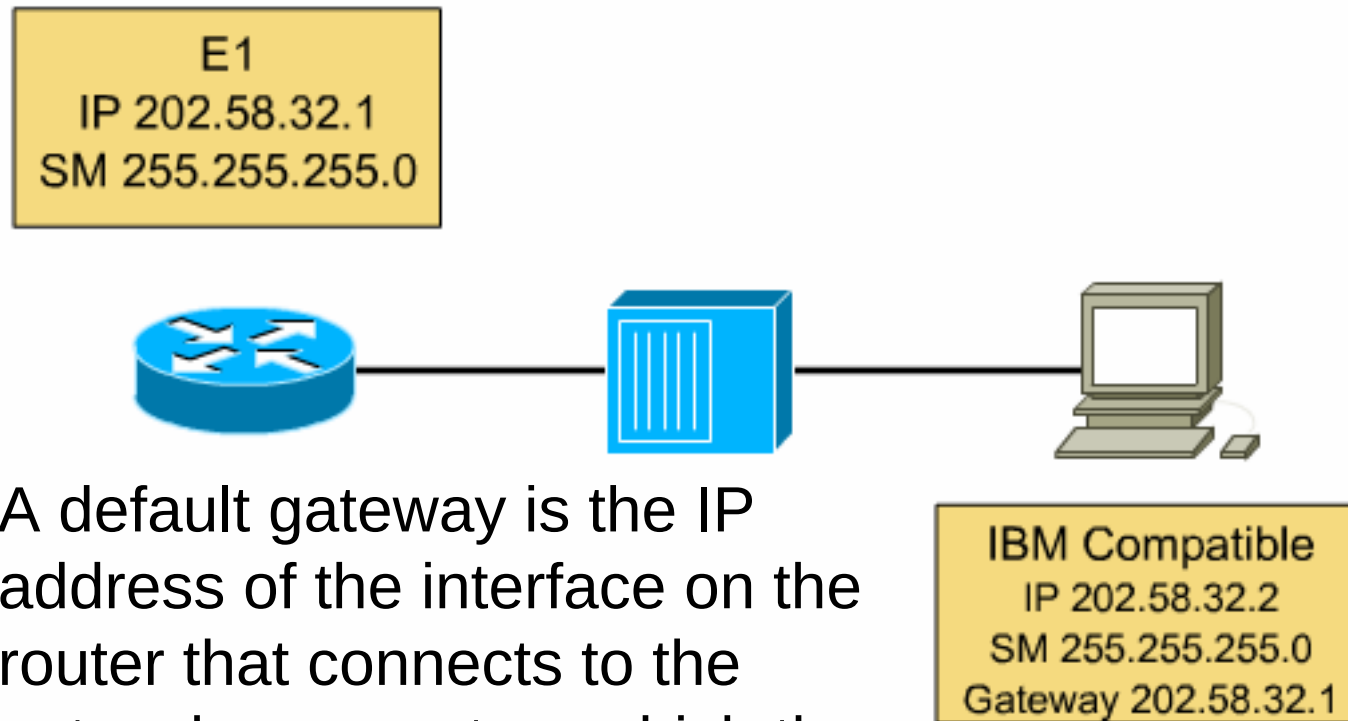
ARP Operation Within a Subnet



ARP Process



Default Gateway



A default gateway is the IP address of the interface on the router that connects to the network segment on which the source host is located.

Important Utilities:

ping -l length 192.168.252.1 ping -n number 192.168.252.1	ping using a given ICMP echo request length send number ICMP echo requests
telnet 192.168.252.1 p	telnet to port p
ifconfig -a netstat -in ifconfig eth0 ip_add net_msk ipconfig /all	List all network interfaces Configure interface eth0 with ip_add and net_msk List all network interfaces (WINDOWS)
netstate -rn route print	Print routing table Print routing table (WINDOWS)
arp -a arp -d arp -s	Show all entries in the ARP table Delete ARP entries Add static ARP entry
tcpdump -n -e -t	Sniff packets and display with numbers, ethernet addresses and time stamp
windump -D windump -I 1	List available interfaces for listening Listen to interface 1
tracert -n 192.168.252.1 tracert -n 192.168.252.1	See the path that the packets follow (numbers only – IP addresses) WINDOWS

References

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- TCP/IP Illustrated, Volume 1, Stevens.
- Java Network Programming and Distributed Computing, Reilly & Reilly.
- Computer Networks: A Systems Approach, Peterson & Davie.
- <http://www.firewall.cx>
- <http://www.javasoft.com>